Power BI Warming Stripes Chart: Step-by-Step Walkthrough

Dataset: GlobalTemp.xlsx (Year, Anomaly), 1880–2025 · Tool: Power BI Desktop

1. Overview

This walkthrough builds a **warming stripes** chart (also called a stripe chart) in Power BI. Each stripe is one year. Every stripe has the same height, and its color shows that year's global temperature anomaly: blue for cooler than the baseline, white near the baseline, and red for warmer.

What you will build

- A clustered column chart where every column has equal height and touches its neighbours
- A diverging blue, white, red color scale driven by a measure
- A legend showing the anomaly range
- A clean report page with a title and short explanations

Data used

Column	Description
Year	Calendar year, 1880 to 2025
Anomaly	Temperature difference from the 1951–1980 average, in °C (range about -0.49 to +1.29)

2. Import the data

1. Open Power BI Desktop and click **Home > Get data > Excel workbook**.
2. Select `GlobalTemp.xlsx`.
3. Tick **Sheet1** and click **Load**.
4. The table is named `Sheet1` by default. Rename it so the formulas below work: in the **Data** pane, right-click the table, choose **Rename**, and type `GlobalTemp`.

5. Open **Table view** and check the types. Year should be a Whole number (or Text) and Anomaly a Decimal number.
6. Select the **Year** column and, on **Column tools**, set **Summarization** to **Don't summarize**. This stops Power BI from adding up the years.

3. Create the measures

Select the `GlobalTemp` table, click **Table tools > New measure**, and enter each formula.

Equal stripe heights

```
Stripe Height = 1
```

Temperature for color

```
Temp Anomaly = AVERAGE(GlobalTemp[Anomaly])
```

Note that the closing parenthesis is required. If you get an error, check that the table name is exactly `GlobalTemp`.

4. Put the measures in their own section

There are two ways to do this.

Option A: a dedicated measures table (recommended)

1. Click **Home > Enter data**. This is in the Home tab, not Column tools.
2. Leave the grid as is and set the table **Name** to `_Measures`. The underscore keeps it at the top of the Fields list.
3. Click **Load**.
4. In the Data pane, click **Stripe Height**. On the **Measure tools** ribbon, set **Home table** to `_Measures`.
5. Repeat for **Temp Anomaly**.
6. Right-click the placeholder `Column1` in `_Measures` and choose **Hide** (or **Delete from model**).

Common mistake: clicking **New column** creates an empty `Column =` inside `GlobalTemp`, with a red warning icon. Cancel it with the red X in the formula bar, then right-click the column and choose **Delete from model**.

Option B: display folders

1. Open **Model view**.
2. Select a measure, and in **Properties** type a **Display folder** name such as `Warming Stripes`.
3. Use the same name for the other measure.

5. Build the chart

1. In **Visualizations**, choose **Clustered column chart**.
2. Drag **Year** to the **X-axis**.
3. Drag **Stripe Height** (from `_Measures`) to the **Y-axis**.
4. Make sure the **Legend** well is empty. Putting a field in it splits each year into its own series and produces thin bars with gaps.
5. Resize the chart to be wide and fairly short.

6. Format the stripes

Open the **Format visual** pane (the paintbrush icon). You can type a keyword in the search box to find settings quickly.

Setting	Where	Value
Space between categories	Columns > Layout	**0%** (this is the "inner padding" setting)
Y-axis	Y-axis	Off
Gridlines	Gridlines	Off
Data labels	Data labels	Off
Axis titles	X-axis > Title, Y-axis > Title	Off

The **Title > Spacing** options only control space around the chart title, so ignore them for this step.

7. Apply the gradient colors

1. Go to **Format visual > Columns > Colors**.
2. Click the **fx** button next to **Color**.
3. Set **Format style** to **Gradient**.
4. Under **What field should we base this on?**, open the dropdown. Temp Anomaly is not in the chart's fields, so look under **All data**, expand `_Measures`, and choose **Temp Anomaly**.
5. Set the three color stops:

Stop	Value setting	Suggested color
Minimum	Lowest value	Blue, `#2EC4FF`
Center	**Number** `0`	White, `#FFFFFF`
Maximum	Highest value	Red, `#FF3B52`

6. Tick **Add a middle color**.
7. Click **OK**.

Why the center must be set to 0: if Center stays on *Middle value*, Power BI places white at the midpoint of your data (about 0.40). Most early years then count as "cooler" and the whole chart looks blue. Setting the center to the number 0 makes the colors mean "below or above the baseline."

8. Show all years and remove the empty space

If the chart shows a scrollbar, or there is blank space on the right, adjust the X-axis.

1. Open **Format visual > X-axis**.
2. To label every 20 years (1880, 1900, 1920…), set **Type** to **Continuous**.
3. Under **Range**, set **Minimum** to `1880` and **Maximum** to `2025` (use `2026` if the last stripe is clipped).
4. Turn **Round range Off**, so Power BI does not extend the axis to a rounder number such as 2050.
5. If gaps appear between stripes with the Continuous type, switch back to **Categorical** and reduce **Minimum category width** until the scrollbar disappears.
6. Make the chart as wide as the page content.

9. Add the legend

Because the colors come from conditional formatting on a measure, Power BI draws the gradient legend itself.

1. Open **Format visual > Legend** and toggle it **On**.
2. Set **Position** to **Top left**.
3. Turn the legend **Title** on (it will read `Temp Anomaly`).
4. Use a small grey font, 9–10 pt.

The legend shows the minimum (-0.49), the center (0.00), and the maximum (+1.29) of your color scale.

Do not build the legend by dragging a category column into the chart's Legend well. That breaks the stripes, as described in step 5. A separate gradient rectangle with text boxes is also possible, but the built-in legend is simpler and stays in sync with the data.

10. Polish the report page

Chart title: turn on **General > Title** and enter `Global temperature 1880–2025 (vs 1951–1980 average)`. Use bold, 12–14 pt, left aligned.

Page and chart background

- Click an empty part of the page, open **Format page > Canvas background**, and choose light grey (`#F7F7F7`) with 0% transparency.
- Select the chart and set **General > Effects > Background** to white, with a thin light-grey border (`#D9D9D9`).

Text boxes (Insert > Text box)

- *Page title:* "Stripe Chart", about 28 pt, bold.
- *What is a Stripe Chart?* A stripe chart shows one value over an ordered sequence, usually time, as thin side-by-side stripes where color encodes the value instead of height.
- *How to Interpret a Stripe Chart?* Read it by looking at color shifts. With a diverging scale, blue means below the baseline, white near average, and red above, so a move from blue to red shows a rising trend. It is best for spotting overall patterns at a glance, not for reading exact values.

Use bold 14 pt for the headings and regular 11–12 pt dark grey (`#333333`) for the paragraphs.

Alignment

1. Select the text boxes and the chart.
2. Use **Format > Align > Align left**.
3. Optionally use **Format > Group** so they move together.

11. Troubleshooting

Problem	Cause	Fix
Formula error on `GlobalTemp[Anomaly]`	Table is still named `Sheet1`	Rename the table to `GlobalTemp`
Years are summed or look wrong	Year is being summarized	Set Summarization to Don't summarize
Cannot find Temp Anomaly in the color dialog	The measure is not in the chart's fields	Choose it under **All data > _Measures**
Everything is blue	Center is on *Middle value*	Set Center to **Number 0**
No red visible	Later years are hidden behind the scrollbar	Widen the chart or fix the X-axis range and minimum category width
Thin bars with gaps and odd colors	A category field is in the Legend well	Remove it from Legend and reapply the gradient
Stripes do not touch	Space between categories above 0%	Set it to 0%
Blank space on the right	Axis auto-rounds to a larger number	Set Maximum to 2025 and turn Round range Off
Empty `Column =` in the table	New column was clicked instead of a new table	Cancel it and delete the column from the model

12. Final checklist

- [ ] Data imported and table renamed to `GlobalTemp`
- [ ] Year set to Don't summarize
- [ ] Measures `Stripe Height` and `Temp Anomaly` created (and moved to `_Measures`)
- [ ] Clustered column chart with Year on X-axis and Stripe Height on Y-axis
- [ ] Space between categories at 0%
- [ ] Y-axis, gridlines, and data labels off
- [ ] Gradient on Temp Anomaly with center at Number 0 (blue, white, red)
- [ ] Legend visible
- [ ] X-axis range 1880–2025, no scrollbar, no blank space
- [ ] Title, background, and explanatory text added